A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2.7 - 2.45 = 0.25 billion = 250 million years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (million years) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans **1** Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Oxygen in the atmosphere absorbed ultraviolet light **1** A dynamic equilibrium of reactions between O, O_2 and O_3 (the ozone layer) developed **1** Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) **1** Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved – many anaerobic Archaea and bacteria died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) **1** The lower CO_2 concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion **1** Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas **1** Regulation of atmospheric composition (eg oxygen / CO_2 by photosynthesis and respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Transpiration** → water lost as vapour from the leaves **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 2 points below. **1 In-situ:** conserving species in their natural habitat with the whole community – ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding **1** Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	2
02.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Is the wild population threatened? **1** Is there a genetically diverse captive population? **1** Is in-situ conservation failing? **1** Is keeping a captive population realistic (habitat size, food, relationships, cost)? **1** Is release likely to be successful now or in future (suitable habitat, threats removed)? **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
02.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Blue whale:** habitat size far too large to provide – food (huge quantities of krill / plankton) cannot be supplied – cost **2 Large blue butterfly:** complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs – requires specific ant species and thyme food plant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Harvesting no faster than the MSY - resource never depleted **1** Mixed species plantations - more food choices, relationships and niches **1** Indigenous species - support indigenous insects, birds **1** Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield **1** Selective logging - small clearings, less disturbance, long-term timber benefit **1** Retention of dead wood, buffer zones, corridors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
04.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Demersal trawl:** seabed species - cod, haddock, plaice, shrimps, scampi – chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass) – high by-catch in mixed fisheries – high fuel use **2 Purse seine:** shoaling pelagic species - tuna, sardines, herring, anchovy – can remove whole shoals – dolphins associated with tuna shoals may be caught **2 Pelagic long-line:** tuna, squid – albatrosses (long-lived, low breeding rate) drown on baited hooks – sharks and turtles caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch **1** Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Lost or discarded nets and traps continue to catch and kill organisms – dead animals act as bait attracting more **1** Biodegradable ropes / panels so lost traps fall apart – radio transmitters on traps so they can be recovered – net recovery schemes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biodegradable** → broken down by decomposers **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.	
05.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Designation of protected areas (SSSIs, MCZs) **1** Protection of wild birds and their nests (except some pests / game birds) **1** Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts **1** Uprooting wild plants generally illegal **1** Bats must not be disturbed – roof timber treatments must be non-toxic to bats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
05.2	**1 mark each** 1 mark for each of the 4 points below. **1 Appendix I:** species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding **1 Examples:** great apes, big cats, rhinos, blue whale **1 Appendix II:** species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected **1 Examples:** Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **CITES** → the treaty that controls international trade in endangered species **Do not credit** • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	4

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market **1** Removing the profit removes the incentive to collect **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	
06.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
06.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** They are driven by renewable, low energy-density resources, especially solar energy, which will not be depleted: the hydrological cycle, the carbon and nitrogen cycles, atmospheric and thermohaline circulation **1** Most natural processes occur at low temperatures, because enzymes lower the activation energy of the reactions, so little energy is needed **1** Processes link into cycles in which the waste product of one is the raw material for the next, so raw materials are not depleted and wastes do not build up **1** A small number of abundant, simple elements are built into monomers and then into a wide range of polymers, so supplies of the raw materials are secure **1** Wastes are biodegradable and break down to non-toxic products, so they do not accumulate or poison the system **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **thermohaline** → the ocean conveyor driven by temperature and salinity • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals are dispersed after use - spread thinly through products all over the world, or lost to corrosion - so there is no concentrated deposit left to recover **1** Products are made of mixtures of materials that are difficult to separate, so recovery and re-use are expensive and often incomplete **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Design for extended use:** a product with a long life, which can be repaired, gives fewer opportunities for its materials to be lost **1 Design for end-of-life re-use:** materials that can be identified and separated easily are far more likely to be recycled **1** Separation of technical and biological materials, because metals and plastics need different reprocessing from biodegradable materials, which can be composted or digested **1** Energy from renewable sources at every stage, as in natural systems **1** Optimum rather than maximum production, so that the system supports the processes it relies on instead of maximising one output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3
07.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sustainability depends on the total impact, which is the number of people multiplied by the impact of each person, so the same activity crosses the threshold as either rises **1** A natural process can absorb or replace what is taken only up to a rate – beyond that the resource is depleted or the waste accumulates faster than it can be broken down **1 Example:** slash and burn agriculture is sustainable where the population is small enough for each clearing to be left fallow long enough to recover, but not when land has to be re-cleared before the soil fertility has returned **1 Example:** discharging untreated sewage into a river is harmless where decomposers can break it down, but causes deoxygenation and eutrophication once the load exceeds that rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **threshold** → the concentration below which no harm is detected • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
07.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Urban populations are supplied by resources brought in from elsewhere, so the damage caused by extraction happens out of sight and often in another country **1** In a sedentary rural community the resource is local, so the problems of over-exploitation become obvious quickly as supplies dwindle, and a cultural belief in limiting exploitation to needs develops **1 Consequence:** when one area is exhausted, exploitation simply moves to another area and the consumer sees no change, so the total damage keeps growing - this has happened with timber, energy, metal minerals and fish **1 Consequence:** there is no pressure from consumers to change, so unsustainable practices continue until the resource fails on a large scale **1 Consequence:** the costs fall on the people and ecosystems where extraction happens, not on those who benefit, so the market price never reflects the true environmental cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-377; AQA spec 3.6.2-3.6.3.	
08.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Selective breeding:** parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness) – combines genes but characteristics of offspring not fully predictable – many generations needed – risk of inbreeding **2** Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high – but no variation so no improvement and vulnerability to the same disease – cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing **2 Cross-breeding:** two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	6

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 4 points below. **1 AI:** semen from a male with desirable characteristics can fertilise many more females than natural mating – frozen semen stored and transported easily and used after the male dies **1 Embryo transfer:** FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male **1** Each embryo is implanted in a different surrogate female **1** So the donor produces many offspring per year rather than one per pregnancy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable **1** If it cannot be adjusted the percentage error should be recorded so the results can be corrected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2
09.2	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1 Droppings:** diet, gender, territory size **1 Owl pellets:** diet **1** Feeding marks (eg chewed hazel nuts): presence of species such as dormice **1 Nests / burrows:** presence and breeding **1 Tracks / footprints:** presence, movements **1** Territorial marks (scratching posts): presence and territory **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species **1** The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Marked individuals mix freely with the rest of the population **1** Marks do not fall off or wear away, and marking does not affect survival **1** The population does not change during the study (no births, deaths or migration) **1** All individuals have an equal chance of being caught / animals are mobile and can be caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Resources:** wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food – new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • **Biomimetics:** vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples) – most species unstudied - argument for habitat conservation • Physiological research using wildlife species – wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools • **Ecosystem services:** atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments – the unknown value of species not yet investigated (precautionary principle) • **Evaluation:** which arguments are most persuasive to decision makers – economic valuation vs intrinsic value – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	9
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Human society depends on other species first as resources: wood, fibres such as cotton, wool and silk, oils, fuels and, above all, food, with wild species such as the morama bean, perennial cereals, eland and bison offering new crops and livestock better adapted to local conditions. Biomimetics copies adaptations into vehicle design, architecture, structures, adhesives, materials and ultrasound; medicines such as taxol, quinine and AZT came from plants and animals, and physiological research on marsupials, hippos and sponges continues to yield medical insights. Wild species are pest-control agents - ladybirds, *Encarsia*, *Cactoblastis*, *Bacillus thuringiensis* - and crop wild relatives in Vavilov centres hold the genes future crops will need.	
	Less visibly, other species provide the services that keep the planet habitable: regulating atmospheric composition, driving the hydrological and biogeochemical cycles, maintaining soil, pollinating crops, dispersing seeds and providing habitat. Most species have never been investigated, so their value is unknown, and there are ethical, aesthetic and recreational reasons for conservation that owe nothing to usefulness.	
	Which arguments persuade decision-makers? Economic valuation of resources and services speaks the language of policy and has been most effective, but it undervalues the services that go unnoticed because they operate in dynamic equilibrium and it can never capture the value of species not yet studied. The strongest case combines the precautionary logic of unknown value with the intrinsic argument: conserving biodiversity protects both what we know we need and what we cannot yet know.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important for humans in many ways. We get food, wood and cotton from plants and animals. Many medicines come from plants. Bees pollinate our crops so we need to conserve them. Animals and plants are also nice to look at and people like to visit nature. We should conserve biodiversity because we might need species in the future.	
	What makes the difference	
	• Level 1 names categories without examples; Level 3 gives specific, accurate examples in every category.	
	• Level 3 includes ecosystem services, genetic resources and pest control that Level 1 omits.	
	• 'Discuss' at the top level means weighing the arguments against each other - Level 3 evaluates which arguments persuade and why; Level 1 does not.	
10.2	1 mark for the point below. **1** The variety and abundance of life (species, genes, habitats) in an area / on Earth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how human activities influence biodiversity, using examples from a range of different contexts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil) – deliberate eradication of predators, competitors, vectors, pests • **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage – inactivity (stopping plagioclimax management) • **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation – loss of pollinators, seed dispersers, keystone species – food chain impacts (sea otters, puffins, jellyfish) • **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs – fragmentation and edge effects • **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica	25
	– agriculture, fishing, forestry • **Positive influences:** habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why monitor:** species present, population size, density, distribution, survival, age structure, breeding rates, movements – baseline and evaluation – Red List categorisation and quotas rely on it • **Standard techniques and their limitations:** quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR – sampling design (random / systematic, size, number, timing, standardisation) • **New technologies:** satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • **How they improve validity:** representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • **Examples:** penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked	25
	• **Limitations:** cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	

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